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Thermodynamic modeling of the Sr–X (X=H, Li, Na, Sc) systems

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ABSTRACT

The Sr–X (X=H, Li, Na, Sc) systems are critically reviewed and modeled by means of the CALPHAD technique. The two-sublattice model is used to describe the solid solution phase in the Sr–H system, and the gas phase is treated as an ideal gas mixture of the species H, H_2, Sr and Sr_2. All the intermetallic compounds (αSrH_2, βSrH_2, Sr_6Li_{23} and Sr_3Li_2) are treated as stoichiometric compounds. By means of first-principles calculations, the enthalpies of formation at 0 K for the Sr_6Li_{23} and Sr_3Li_2 are computed with the desire to assist the thermodynamic modeling. No compound is found in the Sr–Na and Sr–Sc systems. A set of self-consistent thermodynamic parameters is finally obtained for each of these binary systems. Comparisons between the calculated and measured phase diagrams as well as thermodynamic properties show that most of the experimental information can be satisfactorily reproduced by the present thermodynamic descriptions.

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1. Introduction

Al alloys are widely used for commercial applications mainly due to the low density as well as good mechanical properties. H, Li, Na, Sc and Sr are important elements that affect the mechanical properties of Al alloys. Hydrogen embrittlement is considered as an embrittlement mechanism in Al alloys, which appears during the corrosion process [1]. The addition of Li to Al offers the promise of substantially reducing the weight of Al alloys and increasing the performance [2]. Modification of Al alloys with addition of Na or Sr is being applied to further improve the mechanical properties of Al alloys. Na has historically been the predominant modifying agent for Al–Si alloys. However, Sr has substituted for Na in recent years owing to three disadvantages of Na modified Al alloys: difficulty in controlling the concentration, fuming during addition and difficulty in keeping modification effect [3–5]. Sc, which is an alloying element in Al alloys, has extensively received an interest due to the grain refinement during casting or welding, precipitation hardening resulting from Al_3Sc particles and grain structure control from Al_3Sc dispersoids [6]. In the production of Al alloys, knowledge of phase diagrams and thermodynamic properties is essential.

The thermodynamic descriptions for the Sr–X (X=H, Li, Na, Sc) binary systems are a continuing effort of our previous attempts [7–11] to establish a thermodynamic database for multicomponent Al alloys. However, to the best of our knowledge, so far there are no thermodynamic assessments for these binary systems. A thorough thermodynamic assessment of the Sr–X (X=H, Li, Na, Sc) systems is needed in order to provide a reliable basis for thermodynamic extrapolations and calculations in related ternary and higher-order systems. The purpose of the present work is devoted to providing a self-consistent set of thermodynamic parameters for each of these binary systems by means of a hybrid approach of CALPHAD method and first-principles calculations.

2. Literature review

2.1. The Sr–H system

The only experimental phase diagram of pseudobinary Sr–SrH_2 system was determined by Peterson and Colburn [12] by means of differential thermal analysis (DTA), chemical analysis of isothermal equilibration and X-ray diffraction (XRD). The maximum solubility of H in (βSr) was determined to be 43.2 at% H at the peritectic reaction of 880 °C. The αSrH_2, which was stable at low temperature, transformed into βSrH_2 at 855 °C. An intermediate solution phase γ with Hcp_A3 structure was observed to be stable from about 240 to 620 °C.

The enthalpies of formation of αSrH_2 and βSrH_2 shown in Table 1 were determined by Guntz and Benoit [13], Ehrlich et al. [14], Banus and Bragdon [15] and Peterson and Nelson [16], respectively. The data from Peterson and Nelson [16] are used in the present optimization due to the well described experimental procedures. Besides, the value of the enthalpy of formation of αSrH_2 reported in Refs. [13] and [14] agrees well with each

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Table 1
Summary of the experimental enthalpy of formation of SrH_2 in the Sr–H system.

Phase	ΔH (kJ/mol)	Method	Ref.
αSrH_2	−176	Calorimetric method	[13]
	−180	Calorimetric method	[14]
	−197	Evaluation of H_2 equilibrium pressure	[15]
	-183 ± 3	Evaluation of H_2 equilibrium pressure	[16]
βSrH_2	-189 ± 8	Evaluation of H_2 equilibrium pressure	[16]

Table 2
Crystallographic data of solid phases in the Sr−X (X=H, Li, Na, Sc) systems.

Phase/ Temperature Range (°C)	Pearson Symbol/Space Group/Prototype	Lattice Parameters(nm)	Comments/ References
(H)	*hp2*	a=0.3771	25 °C/[45]
< …	$P6_3/mmc$	c=1.631	
	Mg		
(Li)	*cI2*	a=0.35093	25 °C/[45]
< 180.55	$Im\bar{3}m$		
	W		
(Na)	*cI2*	a=0.42096	25 °C/[45]
< 96.95	$Im\bar{3}m$		
	W		
(αSc)	*hp2*	a=0.33088	25 °C/[45]
< 1334.85	$P6_3/mmc$	c=0.52680	
	Mg		
(βSc)	*cI2*	a=0.373	> 1334.85 °C/ [46]
< 1540.85	$Im\bar{3}m$		
	W		
(αSr)	*cF4*	a=0.6084	25 °C/[45]
< 546.85	$Fm\bar{3}m$		
	Cu		
(βSr)	*cI2*	a=0.487	> 546.85 °C/ [46]
< 776.85	$Im\bar{3}m$		
	W		
αSrH_2	*oP12*	a=0.6364	[17]
< 855	*Pnma*	b=0.3875	
	Co_2Si	c=0.7343	
Sr_6Li_{23}	*cF116*	a=1.488	[19]
< 152	$Fm\bar{3}m$		
	Th_6Mn_{23}	a=1.505	[24,25]
Sr_3Li_2	*pT4*	a=0.9628	[19]
< 198	Zr_3Al_2	c=0.855	

other. The crystal structure data of αSrH_2 are taken from [17], as shown in Table 2.

2.2. The Sr–Li system

The Sr–Li system was investigated by Wang et al. [18,19] using DTA and XRD. According to the work by Wang et al. [18,19], the system displayed limited solubility at both ends. Five intermetallic phases were reported: Sr_8Li (Hcp), Sr_8Li (Tet), Sr_7Li, Sr_3Li_2 and Sr_6Li_{23}. The Sr_8Li (Hcp), which was stable at high temperature, transformed into Sr_8Li (Tet.) at 197 ± 1 °C. Three allotropes of Sr were found: αSr (Fcc) up to 231 °C, γSr (Hcp) at 231–623 °C and βSr (Bcc) above 623 °C. The information on different forms of Sr is different from the generally accepted one [20]. Using both DTA and resistance-temperature methods, Hubberstey and Roberts [21] redetermined the Li-rich hypoeutectic liquidus and eutectic points of the Sr–Li system. The eutectic point of reaction Liquid=(Li)+Sr_6Li_{23} was determined to be at 134 °C and 10.1 at% Sr.

Reviewing the work on the SrH_2–Sr system [12], Bale and Pelton [22] pointed out that a hydrogen impurity in the samples of Wang et al. [18] would clearly have a detrimental effect upon the measured Sr–Li phase diagram, particularly on the Sr side. Therefore, the phase relations in the present work on this side are mainly taken from the phase diagram assessed by Bale and Pelton [22]. Bale and Pelton [22] calculated the liquidus mainly using selected experimental liquidus data at the lower Sr concentration regions by Wang et al. [18] and the following approximation equation [23]:

$$(dT/dX_A^L)^{-1}_{x_A=1} - (dT/dX_A^S)^{-1}_{x_A=1} = \Delta h^0_{f(A)}/R(T^0_{f(A)})^2 \quad (1)$$

where $(dT/dX_A^L)^{-1}_{x_A=1}$ and $(dT/dX_A^S)^{-1}_{x_A=1}$ are the slopes of the liquidus and solidus when the mole fraction, X_A, of component A approaches unity, $\Delta h^0_{f(A)}$ is the molar enthalpy of fusion of A, and $T^0_{f(A)}$ is the melting point of A (in Kelvin). Their calculations [22] at Li-rich part were confirmed by Hubberstey and Roberts [21]. As for the Li-rich side, both the experimental data form Wang et al. [18] and Hubberstey and Roberts [21] are utilized in the present work. Since the existence of Sr_8Li and Sr_7Li [18,19] is doubtful and they may be the hydride SrH_2 due to the hydrogen impurity [18,19], none of these compounds are considered in the present work.

There are no thermodynamic data available in this system. The crystal structure data from [19,24,25] are summarized in Table 2.

2.3. The Sr–Na system

The Sr–Na system has been investigated by Remy et al. [26], Roberts [27] and Bussey et al. [28]. A complete miscibility in the liquid state was reported and no intermetallic compound was found. The eutectic point of reaction Liquid=(Na)+(αSr) was reported to be at 92.5 °C and 3.2 at% Sr by Remy et al. [26]. Roberts [27] studied the Sr–Na system using thermal analysis (TA), DTA and XRD over the entire composition range. Three allotropes of Sr were also found. The maximum solubility of Na in (αSr) was observed to be 12.5 at% Na. The solid solubility of Sr in (Na) was very small and could be ignored. The eutectic point was observed to be at 94 °C and 1.3 at% Sr. A peritectic reaction liquid+(βSr)=(γSr) occurred between 83 and 96.5 at% Sr at 635 °C. Another peritectic reaction liquid+(γSr)=(αSr) was in the composition range of 14–90 at% Sr at 192 °C. Later, Bussey et al. [28] redetermined this system up to 37.9 at% Sr and 439 °C using resistivity-temperature, TA and XRD methods. The solubility of Sr in (Na) was less than previously reported one and the eutectic point was at 0.65 at% Sr and 94.8 °C. In the present work, the two peritectic reactions corresponding to the transformations of the three allotropes of Sr [27] are not considered due to the presence of hydrogen. From 0 to 37.9 at% Sr, the liquidus data from Bussey et al. [28] instead of those from Roberts [27] are utilized in view of the difficulty Roberts [27] encountered in detecting thermal arrests for the hypereutectic liquidus. Pelton [23] has also confirmed that the liquidus data of Roberts [27] at Na-rich side are not reliable by using the equation mentioned in Section 2.2.

No thermodynamic data are available in this system.

2.4. The Sr–Sc system

The solubility information of the Sr–Sc system was investigated by Dzhuraev and Altynbaev [29]. The maximum solubility of Sc in (βSr) was determined to be 0.02 at% Sc. An eutectic reaction occurs at 768 °C. On the basis of the above results, and assuming some similarity for the binary systems of Sc or Y with Ca, Ba, or Sr, Moffatt [30] proposed a phase diagram for the Sr–Sc system.

3. Thermodynamic model

3.1. Unary phase

The descriptions of the Gibbs energy of the pure element i (i=Sr, H, Li, Na, Sc) in the phase φ are taken from the SGTE database [20], and the functions are expressed by the following equation:

$$G_i(T)-H_i^{SER}=a+b\times T+c\times T\times \ln T+d\times T^2+e\times T^{-1}$$
$$+f\times T^3+g\times T^7+h\times T^{-9} \quad (2)$$

where H_i^{SER} is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state, and T is the absolute temperature.

3.2. Solution phase

3.2.1. The Sr–X (X=Li, Na, Sc) system

The liquid, (αSr), (Li), (Na), (βSc), (βSr) and (αSc) phases are modeled as completely disordered solutions. The Gibbs energy is described by the Redlich–Kister polynomial [31]

$$G_m^{\phi}-H^{SER}=x_{Sr}\times {}^0G_{Sr}^{\phi}+x_i\times {}^0G_i^{\phi}+R\times T\times (x_{Sr}\times \ln x_{Sr}+x_i\times \ln x_i)$$
$$+x_{Sr}\times x_i\times \sum_{j=0}^{n} L^{j,\phi}(x_{Sr}-x_i)^j \quad (3)$$

where H^{SER} is the abbreviation of $x_{Sr}\times H_{Sr}^{SER}+x_i\times H_i^{SER}$, x_{Sr} and x_i are the mole fractions of Sr and component i (i=Li, Na, Sc), respectively, R is the gas constant. $L^{j,\phi}$ is the jth interaction parameter of phase φ, and it is equal to $L^{j,\phi}=a_j+b_j\times T$. The parameters a_j and b_j are to be optimized in the present work.

Table 3
Summary of the thermodynamic parameters in the Sr–X (X=H, Li, Na, Sc) systems[a].

System	Phase	Model	Evaluated parameters
Sr–H	Liquid	(Sr,H)	${}^0L_{Sr,H}^{Liquid}=9952.2-112.0824T$ ${}^1L_{Sr,H}^{Liquid}=-55206.6$ ${}^2L_{Sr,H}^{Liquid}=-21741.6$
	(αSr)	$(Sr)_1(H,Va)_1$	${}^0G_{Sr:H}^{(\alpha Sr)}=-87514.6+64.6600T+{}^0G_{Sr}^{Fcc_A1}+0.5\times {}^0G_{H_2}^{Gas}$ ${}^0L_{Sr:H,Va}^{(\alpha Sr)}=9037.1$
	(βSr)	$(Sr)_1(H,Va)_3$	${}^0G_{Sr:H}^{(\beta Sr)}=6095.4+38.0943T+{}^0G_{Sr}^{Fcc_A1}+1.5\times {}^0G_{H_2}^{Gas}$ ${}^0L_{Sr:H,Va}^{(\beta Sr)}=-90569.3$
	γ	$(Sr)_1(H,Va)_{0.5}$	${}^0G_{Sr:Va}^{\gamma}=277.1-0.3000T+{}^0G_{Sr}^{Fcc_A1}$ ${}^0G_{Sr:H}^{\gamma}=-28744.4+13.8237T+{}^0G_{Sr}^{Fcc_A1}+0.25\times {}^0G_{H_2}^{Gas}$ ${}^0L_{Sr:H,Va}^{\gamma}=-13407.2+17.9010T$
	αSrH_2	$(Sr)_1(H)_2$	${}^0G_{Sr:H}^{\alpha SrH_2}=-185000.0+132.8400T+{}^0G_{Sr}^{Fcc_A1}+{}^0G_{H_2}^{Gas}$
	βSrH_2	$(Sr)_1(H)_2$	${}^0G_{Sr:H}^{\beta SrH_2}=-182000.0+130.1840T+{}^0G_{Sr}^{Fcc_A1}+{}^0G_{H_2}^{Gas}$
	Gas	(Sr,Sr_2,H,H_2)	${}^0G_{Sr}^{Gas}=R\times T\times \ln(10^{-5}\times P)+F15641T$ ${}^0G_{Sr_2}^{Gas}=R\times T\times \ln(10^{-5}\times P)+F15650T$ ${}^0G_{H}^{Gas}=R\times T\times \ln(10^{-5}\times P)+F10598T$ ${}^0G_{H_2}^{Gas}=R\times T\times \ln(10^{-5}\times P)+F11004T$
Sr–Li	Liquid	(Sr,Li)	${}^0L_{Sr,Li}^{Liquid}=11886.1-22.2135T$
	(αSr)	$(Sr,Li)_1(Va)_1$	${}^0L_{Sr,Li:Va}^{Fcc_A1}=20638.7$
	(βSr), (Li)	$(Sr,Li)_1(Va)_3$	${}^0L_{Sr,Li:Va}^{Bcc_A2}=21590.5$
	Sr_6Li_{23}	$(Sr)_6(Li)_{23}$	${}^0G_{Sr:Li}^{Sr_6Li_{23}}=-9510.8-15.7525T+6\times {}^0G_{Sr}^{Fcc_A1}+23\times {}^0G_{Li}^{Bcc_A2}$
	Sr_3Li_2	$(Sr)_3(Li)_2$	${}^0G_{Sr:Li}^{Sr_3Li_2}=-1891.3-4.0580T+3\times {}^0G_{Sr}^{Fcc_A1}+2\times {}^0G_{Li}^{Bcc_A2}$
Sr–Na	Liquid	(Sr,Na)	${}^0L_{Sr,Na}^{Liquid}=15771.3-9.5063T$
	(αSr)	$(Sr,Na)_1Va_1$	${}^0L_{Sr,Na:Va}^{Fcc_A1}=2021.0+15.5248T$
	(βSr), (Na)	$(Sr,Na)_1Va_3$	${}^0L_{Sr,Na:Va}^{Bcc_A2}=14837.8$ ${}^1L_{Sr,Na:Va}^{Bcc_A2}=2266.1$
Sr–Sc	Liquid	(Sr,Sc)	${}^0L_{Sr,Sc}^{Liquid}=56436.8$ ${}^1L_{Sr,Sc}^{Liquid}=-7593.7$
	(αSr)	$(Sr,Sc)_1Va_1$	${}^0L_{Sr,Sc:Va}^{Fcc_A1}=28832.1$
	(βSr), (βSc)	$(Sr,Sc)_1Va_3$	${}^0L_{Sr,Sc:Va}^{Bcc_A2}=71613.1$
	(αSc)	$(Sr,Sc)_1Va_{0.5}$	${}^0L_{Sr,Sc:Va}^{Hcp_A3}=103086.8$

[a] Gibbs energy in J/mol, temperature (T) in Kelvin and pressure (P) in Pa. The Gibbs energies for pure elements are from the SGTE compilation [20], and the Gibbs energies for gas species are from [33].

3.2.2. The Sr–H system

For the liquid phase, the description of the Gibbs energy is identical as Eq. (3). All the solution phases ((αSr), (βSr), γ) can dissolve H. The Gibbs energy functions of these phases are described using two-sublattice model: $(Sr)_1(H, Va)_1$ for (αSr), $(Sr)_1(H, Va)_3$ for (βSr) and $(Sr)_1(H, Va)_{0.5}$ for γ, where hydrogen is assumed to occupy the interstitial sites. Taking the γ phase as example, its molar Gibbs energy can be expressed as

$$G_m^{\gamma} = y_H^2 \times {}^0G_{Sr:H}^{\gamma} + y_{Va}^2 \times {}^0G_{Sr:Va}^{\gamma} + R \times T \times [y_H^2 \times \ln(y_H^2) + y_{Va}^2 \times \ln(y_{Va}^2)] + y_H^2 \times y_{Va}^2 \times {}^0L_{Sr:H,Va}^{\gamma} + y_H^2 \times y_{Va}^2 \times {}^0L_{Sr:H,Va}^{\gamma} \times (y_H^2 - y_{Va}^2) + \ldots \quad (4)$$

$${}^0G_{Sr:H}^{\gamma} = A' + B' \times T + {}^0G_{Sr}^{Fcc_A1} + 0.25^{*0}G_{H_2}^{Gas} \quad (5)$$

$${}^0G_{Sr:Va}^{\gamma} = A'' + B'' \times T + {}^0G_{Sr}^{Fcc_A1} \quad (6)$$

$${}^0L_{Sr:H,Va}^{\gamma} = a_0 + b_0 \times T \quad (7)$$

where ${}^0G_{Sr:H}^{\gamma}$ and ${}^0G_{Sr:Va}^{\gamma}$ are the Gibbs energies of the "end-members", y_i^j is the site fraction of component i in sublattice j, and ${}^0L_{Sr:H,Va}^{\gamma}$ is the interaction parameter. The parameters A', B', A'', B'', a_0 and b_0 are to be optimized.

3.3. Intermetallic compounds

All the four compounds (αSrH_2, βSrH_2, Sr_6Li_{23} and Sr_3Li_2) are treated as stoichiometric compounds. The Gibbs energy of αSrH_2, for example, is expressed by the following equation, adopting the stable structure for the elements as the thermodynamic standard state:

$$G_m^{\alpha SrH_2} - H_{Sr}^{SER} - 2H_H^{SER} = a^{\alpha SrH_2} + b^{\alpha SrH_2} \times T + {}^0G_{Sr}^{Fcc_A1} + {}^0G_{H_2}^{GAS} \quad (8)$$

3.4. Gas phase

Hydrogen deviates from the ideal gas behavior at very high pressures, but it can be approximated as an ideal gas under normal conditions (T=298.15 K, P=1 atom) [32]. The gas phase in the Sr–H system is then modeled as an ideal gas mixture of the species Sr, Sr_2, H and H_2, and the interaction among various species was neglected. Its molar Gibbs energy is given by the following expressions:

$$G^{Gas} - H^{Gas} = \sum y_i[G_i^{Gas} - H_i^{SER} + R \times T \times \ln(y_i)] + R \times T \times \ln(P/\text{bar}) \quad (9)$$

$$(n_{Sr}^0 + n_H^0)/n = y_{Sr} + 2y_{Sr_2} + y_H + 2y_{H_2} \quad (10)$$

where n is the number of moles of the species in the gas in internal equilibrium, $(n_{Sr}^0 + n_H^0)$ is the number of moles of atoms in the gas, $y_i = (n_i/n)$ is the mole fraction of the species i, $G^{Gas} - H^{Gas}$ is the Gibbs energy of the species i, and P is the pressure. The Gibbs energy functions of the individual gas species are taken from the SGTE substance database [33].

4. First-principles calculation

In the present work, first-principles calculations based on density functional theory (DFT) [34] are carried out to evaluate the enthalpies of formation of Sr_6Li_{23} and Sr_3Li_2 in the Sr–Li system, which employ the plane-wave pseudopotential total energy method and are implemented in Vienna ab initio simulation package (VASP) [35,36]. Herein, the calculations adopt projector augmented wave potentials [37] and the generalized gradient approximation of Perdew–Burke–Ernzerhof [38] to depict ion–electron interaction and exchange-correlation function, respectively. For consistency, the cutoff energy is selected to be 350 eV for all calculations, and the samplings of k-points are generated according to Monk–Horst Mesh [39] are 4*4*4, 9*9*11 for Sr_6Li_{23} and Sr_3Li_2, respectively. The convergence criterion of electron self-consistency is chosen as 10^{-6} eV/atom. All structures are fully relaxed with the Methfessel–Paxton method [40], and calculations with the tetrahedron method with Blöch corrections [41] for the integration in reciprocal space are performed.

The enthalpy of formation of the Sr_aLi_b compound is defined as follows:

$$\Delta_f E(Sr_aLi_b) = E(Sr_aLi_b) - \frac{a}{a+b}E(Sr) - \frac{b}{a+b}E(Li) \quad (11)$$

where $E(Sr_aLi_b)$ is equilibrium energy of compound Sr_aLi_b, and $E(Sr)$, $E(Li)$ are the equilibrium energy of pure elements Sr and Li at their standard reference states. Since the present calculations are performed at 0 K and 0 Pa, the enthalpies of formation are equal to the energy of formation. Neglecting the effect of temperature,

Table 4
Calculated invariant equilibria in the Sr–X (X=H, Li, Na, Sc) systems compared with literature data.

Systems	Composition (at% X)			T (°C)	Reference
Sr–H	Liquid+βSrH_2=(βSr)				
	–	66.67	43.2	880	[12]
	40.07	66.67	43.36	885.3	This work
	(βSr)+βSrH_2=αSrH_2				
	–	66.67	66.67	855	[12]
	40.69	66.67	66.67	856.4	This work
	(βSr)+αSrH_2=γ				
	–	66.67	21.90	620	[12]
	17.40	66.673	21.46	624.8	This work
	(βSr)+γ=(αSr)				
	–	–	–	–	–
	1.19	1.68	1.27	549.9	This work
	γ=(αSr)+αSrH_2				
	–	–	66.67	240	[12]
	5.83	5.46	66.67	241.9	This work
Sr–Li	Liquid=(Li)+Sr_6Li_{23}				
	89.90	–	79.31	134	[21]
	91.08	99.93	79.31	135.7	This work
	Liquid+Sr_3Li_2=Sr_6Li_{23}				
	80.00	40.00	79.31	152	[18]
	80.22	40.00	79.31	149.3	This work
	Liquid+(αSr)=Sr_3Li_2				
	60.00	–	40.00	198	[18]
	59.79	0.28	40.00	196.7	This work
	Liquid+(βSr)=(αSr)				
	–	–	–	–	–
	16.54	0.26	0.26	546.8	This work
Sr–Na	Liquid=(Na)+(αSr)				
	96.80	–	–	92.5	[26]
	98.70	–	–	94	[27]
	99.35	–	12.40	94.8	[28]
	99.41	99.71	12.60	96.4	This work
	(βSr)=Liquid+(αSr)				
	–	–	–	–	–
	7.58	55.43	6.02	459.0	This work
Sr–Sc	Liquid#1=(βSc)+Liquid#2				
	96.50	98.00	1.40	1472	[29]
	95.81	99.51	1.37	1480.7	This work
	(βSc)=Liquid#2+(αSc)				
	–	–	–	–	–
	99.68	0.76	99.97	1319.3	This work
	Liquid#2=(βSr)+(αSc)				
	0.04	0.02	98.4	768	[29]
	0.03	0.02	100	776.8	This work
	(βSr)=(αSc)+(αSr)				
	–	–	–	≈547	[29]
	0	100	0.01	547.2	This work

the enthalpies of formation are approximately regarded as the values at 298 K. The calculated enthalpies of formation of Sr_6Li_{23} and Sr_3Li_2 are −9.5108 and −1.8913 kJ/mol, respectively. These values are fixed in the present thermodynamic optimization.

5. Results and discussion

The optimization of the thermodynamic parameters in the Sr–*X* (*X*=H, Li, Na, Sc) systems was performed with the PARROT module of the Thermo-calc software [42]. The step-by-step optimization procedure described by Du et al. [43] was adopted in the present assessment.

Taking the Sr–H system as an example, the optimization began with the βSrH_2 phase, and $a^{\beta SrH_2}$ was fixed according to the experimental enthalpy of formation [16]. In view of the experimental data associated with the liquid phase and the invariant reaction Liquid+βSrH_2=(βSr), the parameters for the liquid phase a_0, b_0, a_1 and a_2 were introduced. The value of $b^{\beta SrH_2}$ was then optimized by using the phase diagram data associated with liquid phase and βSrH_2. $a^{\alpha SrH_2}$ in Eq. (8) was fixed according to the experimental enthalpy of formation [16]. $b^{\alpha SrH_2}$ was obtained on the basis of the allotropy transformation from βSrH_2 to αSrH_2, and one parameter A' in Eq. (5) of (βSr) was used simultaneously. Subsequently the optimization went on by considering the rest invariant reactions. Each time when the invariant reaction was calculated, the parameters of the involved phases in the reaction were optimized, and the solidus data were also considered at the same time. Finally, the thermodynamic parameters for all the phases were optimized simultaneously by considering all the available experimental data. The optimization of the other three systems was analogy with that for the Sr–H system. Attention should be paid that an inverse miscibility gap of liquid phase occurred at high temperatures during the optimization of the Sr–H system. To avoid this, it was necessary to guarantee $\partial^2 G/\partial x^2 > 0$ for the whole phase region [44]. In the new version of the thermo-calc software, a quantity *QF* (*liquid*), which has the identical sign to $\partial^2 G/\partial x^2$, was introduced to meet the constraint.

The thermodynamic parameters are summarized in Table 3 and the calculated invariant temperatures and concentrations along with the experimental ones are listed in Table 4.

The calculated Sr–H phase diagram including gas phase together with the experimental data is shown in Fig. 1(a). As can be seen from this figure, the experimental data are well reproduced by the modeling. The difference between the calculated and the measured temperatures is less than 6 °C. The enlarged phase diagram in the composition range of 0–8 at% H from 200 to 600 °C is presented in Fig. 1(b). In Fig. 2, the calculated enthalpy of formation of the compound αSrH_2 is compared with the experimental ones by Guntz and Benoit [13], Ehrlich et al. [14], Banus and Bragdon [15] and Peterson and Nelson [16]. A satisfactory agreement is observed between the measured and calculated values.

Fig. 3 shows the calculated Sr–Li phase diagram with the experimental data of [21]. The calculated eutectic point of reaction Liquid=(Li)+Sr_6Li_{23} lies at 135.7 °C, 8.92 at% Sr, while the experimental one lies at 134 °C, 10.1 at% Sr [21]. The comparisons in Fig. 3 and Table 4 also show that the present calculation can well describe the experiment data. The difference between the calculated temperatures and the measured ones is less than 3 °C. The enthalpies of formation of the compounds Sr_6Li_{23} and Sr_3Li_2 in the present optimization are taken to be equal to the first-principles calculations results.

Fig. 4 presents the calculated Sr–Na phase diagram along with the experimental data [27,28]. The calculated eutectic reaction of Liquid=(Na)+(αSr) is located at 96.4 °C and 0.59 at% Sr, and the calculation is in good agreement with

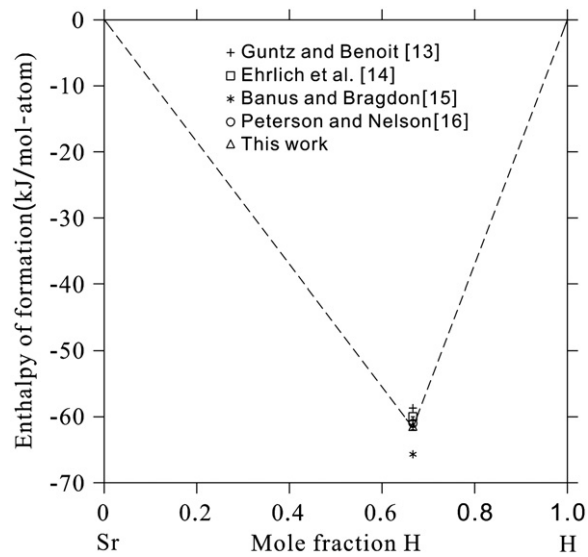

Fig. 2. Calculated enthalpy of formation of the Sr–H system at 298 K compared with the experimental data [13–16].

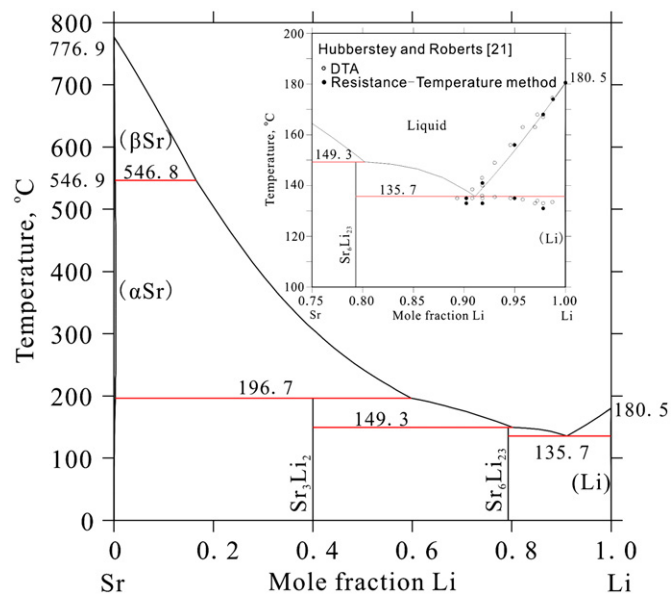

Fig. 3. Calculated Sr–Li phase diagram with the experimental data [21].

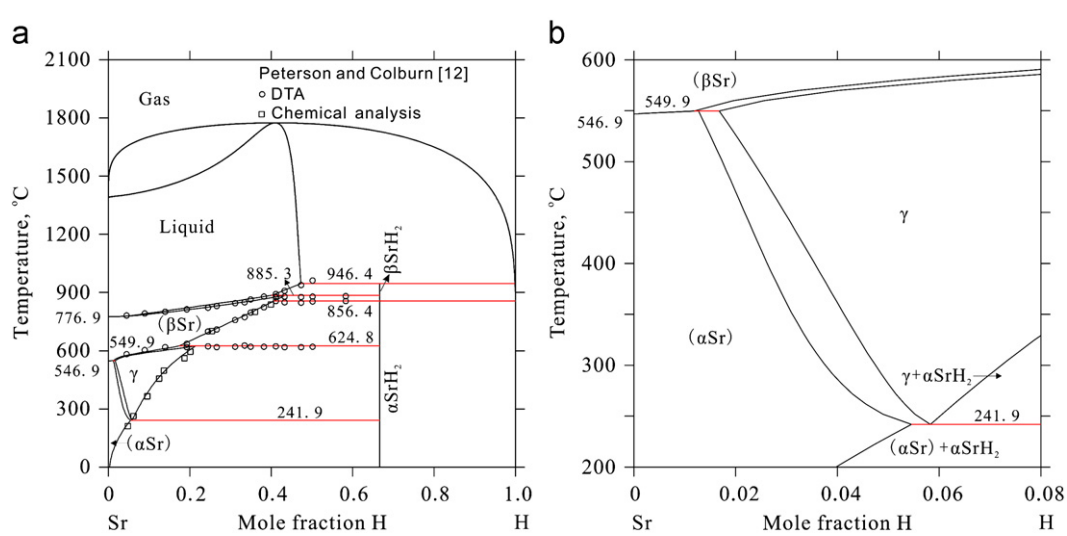

Fig. 1. (a) Calculated Sr–H phase diagram according to the present optimization with the experimental data [12]. (b) Enlarge phase diagram at Sr-rich side.

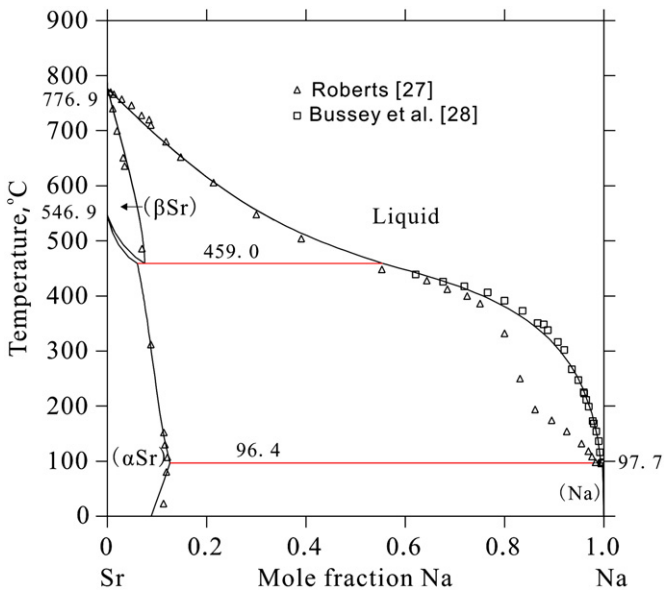

Fig. 4. Calculated Sr–Na phase diagram with the experimental data from [27,28].

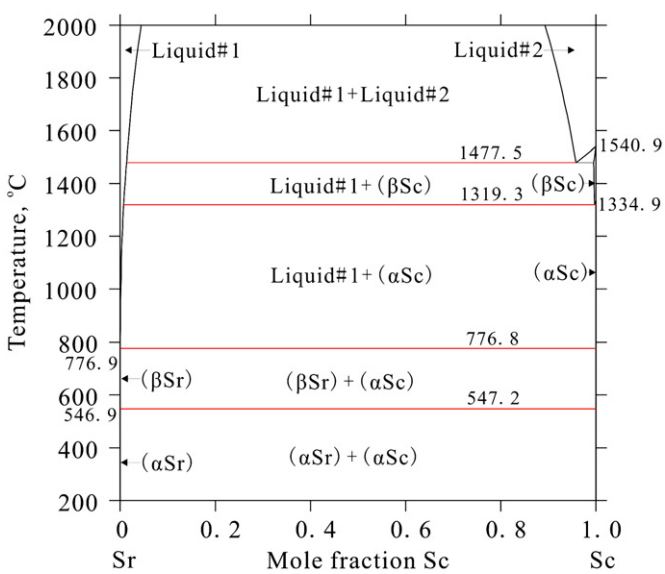

Fig. 5. Calculated Sr–Sc phase diagram according to the present optimization.

the experimental data [28]. The computed liquids and solidus also accord well with the experimental data.

The calculated Sr–Sc phase diagram is shown in Fig. 5. The phase diagram proposed by Moffatt [30] is reproduced by the presented calculation. According to the calculation, a miscibility gap of the liquid phase appears at 1477.5 °C.

6. Conclusion

- A critical literature review of the Sr–X (X=H, Li, Na, Sc) systems has been made to the experimental phase diagram and thermodynamic data. The enthalpies of formation for the two compounds Sr_6Li_{23} and Sr_3Li_2 computed via first-principles calculations are used in the present thermodynamic modeling.
- A set of self-consistent thermodynamic parameters is obtained by the CALPHAD approach for the Sr–X (X=H, Li, Na, Sc) systems. The comprehensive comparisons show that the calculated phase diagram and thermodynamic properties are in a good agreement with the experimental data and first-principles calculations.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2012.03.001.

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